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Study of helicopter autorotation landing following engine failure based on a six-degree-of-freedom rigid-body dynamic model

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Abstract This paper focuses on the prediction of the safe autorotation landing operations of a helicopter following engine failure. The autorotation landing procedure is formulated as a nonlinear optimal control problem based on an augmented six-degree-of-freedom rigid-body flight dynamic model. First, the cost function and constraints are properly selected. The direct transcription approach is then employed to solve the optimal control problem. For a UH-60 helicopter, the optimal solutions with the rigid-body model are compared with those obtained using a two-dimensional point-mass model. It is found that the optimal solutions using the two different models show reasonably good agreement, and furthermore the optimal solutions using the rigid-body model involve the time histories of angular rates and attitudes, lateral velocity and position, as well as pitch controls. Finally the optimal control formulations with different cost functions are proposed for taking account of 1-s time delay and minimum touchdown speed. The calculated control strategies and trajectories are realistic.

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1. Introduction

According to statistics, 50% of helicopter crashes are due to engine failure and there is one engine failure in every 10,000 h of operation. Therefore, of primary concern is how

to operate helicopter landing safely after engine failure. Autorotation landing is usually used by helicopter pilots in case of engine failure. Safety of the landing depends on various factors such as the ambient conditions, the gross weight, and the skill of the pilot. Flight tests are the ultimate methods of checking the safety. However, these tests are very risky, time-consuming, and expensive to conduct. To reduce risks and costs, numerical optimization methods have been proposed to predict optimal safe landing procedures to provide a benchmark of optimal maneuvers for flight tests.

A number of works studied numerical optimization of helicopter autorotation landing after engine failure with two-dimensional point-mass models, but few with six-degree-of-freedom rigid-body models.^{1–10} Johnson used nonlinear

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optimal control theory to study the autorotation landing of a helicopter in hover.¹ Lee et al. extended the work of Johnson by the addition of control and state inequality constraints.^{2,3} Floros modified and extended the works of Johnson and Lee et al. further by using thrust coefficient and disk angle as controls instead of the horizontal and vertical components of thrust coefficient.⁴ They employed two-dimensional point-mass models and found the collective pitch and rotor speed time histories of the optimal solutions in good agreement with those from flight tests. Chen and Zhao investigated the optimal trajectories for a twin-engine helicopter in one-engine-inoperative vertical-takeoff-and-landing and short-takeoff-and-landing terminal-area operations.⁵ To avoid discontinuity of the thrust coefficient at the point of engine failure, an augmented two-dimensional point-mass model was used for the trajectory optimization study. Jhemi studied optimal flight of a helicopter after engine failure with a two-dimensional point-mass model and introduced a direct transcription method in conjunction with a nonlinear programming algorithm to solve the optimization problem.⁶ Aponso et al. used a two-dimensional point-mass model and a real-time trajectory optimization method to show the capability of providing helicopter pilots with autorotation guidance and training.^{7,8} The point-mass models, however, could not involve the motions and constraints of helicopter attitudes, of which the pitch attitude in particular plays an important role in a pilot's view and safe landing. Okuno et al. applied nonlinear optimal control theory to a rigid-body model with longitudinal three degrees of freedom to predict the height-velocity diagram and to study the optimal landing procedures for given initial flight conditions.¹¹ The rotor speed and pitch attitude time histories of the optimal solutions are compatible with those of the flight test data. However, there are jump discontinuities of collective pitch and cyclic pitch histories which are unrealistic. Besides, the cyclic pitch history has kind of bang-bang type, which is different from that of the flight test data. Bottasso et al. studied the computational procedures for rotorcraft trajectory optimization to ensure flyable trajectories and realistic control strategies based on a longitudinal three-degree-of-freedom rigid-body model.^{12,13} A six-degree-of-freedom rigid-body model is

a more comprehensive model of defining helicopter maneuvering flight and able to provide more valuable information on flight trajectory and control strategy. Bibik et al. studied the optimal control of a helicopter after power failure using a linear-quadratic control approach based on a six-degree-of-freedom rigid-body model.^{14,15} However, the computed control inputs are inadequate, because of the high frequency content and unrealistically high control time rates.

In this paper, an augmented six-degree-of-freedom rigid-body flight dynamic model of a helicopter, described in terms of a set of nonlinear differential-algebraic equations, is used for this trajectory optimization study. Considering safety-related requirements and helicopter performance limits, the trajectory optimization problem of helicopter autorotation landing is then formulated into a nonlinear optimal control problem. Finally, optimal solutions are obtained by solving the optimal control problem using nonlinear programming.

2. Dynamic model of a helicopter

A basic three-dimensional rigid-body flight dynamic model of a helicopter is given, with the following six degrees of freedom: longitudinal, lateral, vertical translational motions, and roll, pitch, yaw angular motions. The state variables are velocity components of the helicopter's center of gravity in the body axes (u, v, w), roll, pitch, yaw rates (p, q, r), roll, pitch, yaw attitudes (ϕ, θ, ψ), longitudinal, lateral, vertical positions (x, y, h), main rotor rotational speed Ω , and available shaft power P_S . The control variables are main rotor collective pitch θ_0 , lateral cyclic pitch θ_c , longitudinal cyclic pitch θ_s , and tail rotor collective pitch θ_t . The governing equations are derived and summarized below: where m is the helicopter mass; ($I_{xx}, I_{yy}, I_{zz}, I_{xz}$) are the nonzero components of the helicopter's tensor of inertia; ($F_x, F_y, F_z, M_x, M_y, M_z$) are the combined external forces and moments in the body axes; P_{mr}, P_{tr} are the power required by the main rotor and the tail rotor; η is the helicopter power efficient factor; I_R, I_{tR} are the polar moments of inertia of the main rotor and the tail rotor; k is the ratio of nominal tail rotor angular speed to nominal main rotor angular speed; t_p is

$$\begin{cases} \dot{u} = rv - qw + F_x/m \\ \dot{v} = pw - ru + F_y/m \\ \dot{w} = qu - pv + F_z/m \\ \dot{p} = [(I_{xx} - I_{yy} + I_{zz})I_{xz}pq - (I_{zz}^2 + I_{xz}^2 - I_{yy}I_{zz})qr + I_{zz}M_x + I_{xz}M_z]/(I_{xx}I_{zz} - I_{xz}^2) \\ \dot{q} = [(I_{zz} - I_{xx})pr - I_{xz}(p^2 - r^2) + M_y]/I_{yy} \\ \dot{r} = [(I_{xx}^2 + I_{zz}^2 - I_{xx}I_{yy})pq - (I_{xx} - I_{yy} + I_{zz})I_{xz}qr + I_{xz}M_x + I_{xx}M_z]/(I_{xx}I_{zz} - I_{xz}^2) \\ \dot{\phi} = p + q \sin \phi \tan \theta + r \cos \phi \tan \theta \\ \dot{\theta} = q \cos \phi - r \sin \phi \\ \dot{\psi} = q \sin \phi \sec \theta + r \cos \phi \sec \theta \\ \dot{x} = u \cos \theta \cos \psi + v(\sin \phi \sin \theta \cos \psi - \cos \phi \sin \psi) + w(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \\ \dot{y} = u \cos \theta \sin \psi + v(\cos \phi \cos \psi + \sin \phi \sin \theta \sin \psi) + w(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \\ \dot{h} = u \sin \theta - v \sin \phi \cos \theta - w \cos \phi \cos \theta \\ \dot{\Omega} = \frac{1}{(I_R + k^2 I_{tR})\Omega} \left[P_S - \frac{1}{\eta} (P_{mr} + P_{tr}) \right] \\ \dot{P}_S = \frac{1}{t_p} (P_{OEI} - P_S) \end{cases} \quad (1)$$

the turboshaft engine time constant, and P_{OEI} is the maximum one-engine-inoperative (OEI) power available.

The required power coefficient of the main rotor C_{pm} is calculated from the following equation^{16,17}

$$C_{pm} = -C_T\mu_z - C_H\mu_x + C_S\mu_y + K_{ind}f_G C_T \sqrt{\frac{C_T}{2}} \bar{v}_i + \frac{1}{8} \sigma C_D (1 + 4.7\mu^2)$$

where C_T , C_H , C_S are the main rotor thrust coefficient, drag force coefficient, and side force coefficient; (μ_x, μ_y, μ_z) are the dimensionless velocity components of the main rotor hub in the shaft axes; K_{ind} is the induced power factor of the main rotor; C_D is the mean profile drag coefficient of main rotor blades; σ is the main rotor solidity ratio, and μ is the main rotor advance ratio. The normalized induced velocity of the main rotor is computed using^{1,2}

$$\bar{v}_i = \begin{cases} \frac{1}{\sqrt{\bar{U}_t^2 + (\bar{U}_c + \bar{v}_i)^2}} & (2\bar{U}_c + 3)^2 + \bar{U}_t^2 > 1 \\ \bar{U}_c(0.373\bar{U}_c^2 + 0.598\bar{U}_t^2 - 1.991) & \text{otherwise} \end{cases}$$

where

$$\begin{cases} \bar{U}_c = \frac{-\mu_z}{\sqrt{C_T/2}} \\ \bar{U}_t = \sqrt{\frac{\mu_x^2 + \mu_y^2}{C_T/2}} \end{cases}$$

From Ref. ¹⁸, the ground effect factor f_G is given by the following expression

$$f_G = 1 - \frac{\sigma a \lambda}{4C_T} \cdot \frac{(R/4z)^2}{1 + (\mu/\lambda)^2}$$

where R is the main rotor radius; a is the main rotor blade lift-curve slope; λ is the main rotor inflow ratio, and z is the main rotor hub elevation above ground. The induced velocity of the main rotor is given by¹

$$v_i = K_{ind} \bar{v}_i f_G \cdot \Omega R \sqrt{C_T/2}$$

To express the required power and induced velocity of the main rotor conveniently, C_T is used as an algebraic variable, and the corresponding algebraic equation of C_T is¹⁷

$$C_T - \frac{1}{2} \kappa a \sigma \left[\left(\frac{1}{3} + \frac{1}{2} (\mu_x^2 + \mu_y^2) \right) \theta_0 + \frac{1}{4} (1 + \mu_x^2 + \mu_y^2) \theta_1 - \frac{1}{2} \lambda - \frac{1}{2} \mu_y \theta_c - \frac{1}{2} \mu_x \theta_s + \frac{1}{4} \mu_x \omega_x + \frac{1}{4} \mu_y \omega_y \right] = 0 \quad (2a)$$

where κ is the tip-loss factor of main rotor blades; θ_1 is the main rotor blade twist, and $(\omega_x, \omega_y, \omega_z)$ is the dimensionless angular velocity components of the helicopter in the shaft axes. In a similar way, the algebraic equation of the tail rotor thrust coefficient C_t is incorporated¹⁷

$$C_t - \frac{1}{2} a_t \sigma_t \left[\left(\frac{1}{3} + \frac{1}{2} \mu_t^2 \right) \theta_t + \frac{1}{4} (1 + \mu_t^2) \theta_{1t} - \frac{1}{2} \lambda_t \right] = 0 \quad (2b)$$

where a_t is the tail rotor blade lift-curve slope; σ_t is the tail rotor solidity ratio; θ_{1t} is the tail rotor blade twist; μ_t is the tail rotor advance ratio, and λ_t is the main rotor inflow ratio. The differential-algebraic equations composed of Eqs. (1)

and (2), thus describe the basic six-degree-of-freedom rigid-body flight dynamic model.

To account for the limits on control rates and avoid jump discontinuities of controls, the time derivatives of θ_0 , θ_c , θ_s , and θ_t , instead of θ_0 , θ_c , θ_s , and θ_t themselves, are used as control variables, denoted by u_0 , u_c , u_s , and u_t . The corresponding differential equations are

$$\dot{\theta}_0 = u_0, \quad \dot{\theta}_c = u_c, \quad \dot{\theta}_s = u_s, \quad \text{and} \quad \dot{\theta}_t = u_t \quad (3)$$

The augmented six-degree-of-freedom rigid-body flight dynamic model is described as a semi-explicit index-one system of differential-algebraic equations, consisting of Eqs. (1)–(3). In this augmented rigid-body model, Eqs. (1) and (3) are system differential equations, and Eq. (2) is path equality constraints of the optimal control problem, with differential states u , v , w , p , q , r , ϕ , θ , ψ , x , y , h , Ω , P_S , θ_0 , θ_c , θ_s , θ_t , algebraic states C_T , C_t , and control variables u_0 , u_c , u_s , u_t .

3. Formulation of optimal control problem

The helicopter is assumed to be in steady flight at the time of one-engine failure with rotor speed Ω_0 , forward speed V_0 , height h_0 , flight-path angle γ_0 , and without sideways motion. The trajectory optimization problem is to arrive at the ground with small vertical and horizontal velocities subject to maintaining acceptable conditions during the autorotation landing with 1-s constant-control pilot delay. The optimal control problem can therefore be formulated to minimize the velocity at touchdown, subject to the equations of motion, initial conditions, path constraints, and terminal constraints. The cost function can be chosen as below^{10,11}

$$\min J^p = \left[\left(\frac{u(t_f)}{u_{fmax}} \right)^2 + \left(\frac{w(t_f)}{w_{fmax}} \right)^2 \right]$$

where t_f is the unspecified terminal time of the optimal control problem; u_{fmax} and w_{fmax} are the horizontal and vertical components of the maximum allowable touchdown speed. The equations of motion are Eqs. (1) and (3). In addition to Eq. (2), the following path constraints are considered

$$\begin{cases} u_{min} \leq u(t) \leq u_{max} & v_{min} \leq v(t) \leq v_{max} \\ w_{min} \leq w(t) \leq w_{max} & p_{min} \leq p(t) \leq p_{max} \\ q_{min} \leq q(t) \leq q_{max} & r_{min} \leq r(t) \leq r_{max} \\ \phi_{min} \leq \phi(t) \leq \phi_{max} & \theta_{min} \leq \theta(t) \leq \theta_{max} \\ \psi_{min} \leq \psi(t) \leq \psi_{max} & x_{min} \leq x(t) \leq x_{max} \\ y_{min} \leq y(t) \leq y_{max} & h(t) \geq 0 \\ \Omega_{min} \leq \Omega(t) \leq \Omega_{max} & P_{Smin} \leq P_S(t) \leq P_{Smax} \\ C_{Tmin} \leq C_T(t) \leq C_{Tmax} & C_{min} \leq C_t(t) \leq C_{max} \\ \theta_{0min} \leq \theta_0(t) \leq \theta_{0max} & \theta_{cmin} \leq \theta_c(t) \leq \theta_{cmax} \\ \theta_{smin} \leq \theta_s(t) \leq \theta_{smax} & \theta_{tmin} \leq \theta_t(t) \leq \theta_{tmax} \\ u_{0min} \leq u_0(t) \leq u_{0max} & u_{cmin} \leq u_c(t) \leq u_{cmax} \\ u_{smin} \leq u_s(t) \leq u_{smax} & u_{tmin} \leq u_t(t) \leq u_{tmax} \end{cases}$$

The initial conditions are determined by the flight conditions of the helicopter at the time of initial control actuation after engine failure, i.e., the initial time of the optimal control problem t_0 , equal to pilot delay time of 1 s. The corresponding values of the state variables at time t_0 are obtained numerically by solving the initial-value problem of differential-algebraic

equations composed of Eqs. (1) and (2) with controls fixed to the initial trim values, using the backward differentiation formulas. The terminal constraints are formulated as follows:

$$\begin{cases} 0 \leq u(t_f) \leq u_{\max} & v(t_f) = 0 \\ 0 \leq w(t_f) \leq w_{\max} & p(t_f) = 0 \\ q(t_f) = 0 & r(t_f) = 0 \\ \phi(t_f) = 0 & \theta_{\min} \leq \theta(t_f) \leq \theta_{\max} \\ \psi(t_f) = 0 & y(t_f) = 0 \\ h(t_f) = 0 \end{cases}$$

For the nonlinear optimal control problem, in addition to the different units used, the state and control variables differ in the range of magnitude by great amounts. The computational efficiency and the rate of convergence of numerical optimization methods, however, depend critically on the scales used for the variables, especially in nonlinear problems. Therefore, all the variables need to be normalized and scaled to make them have the same range, e.g., [0, 1]. Nondimensionalized and scaled quantities for the state and control variables as well as the time used in the analysis are defined as follows:

$$\begin{cases} \bar{u} = \frac{1}{k_v \Omega_0 R} u & \bar{v} = \frac{1}{k_v \Omega_0 R} v & \bar{w} = \frac{1}{k_v \Omega_0 R} w \\ \bar{p} = \frac{k_x}{k_v \Omega_0} p & \bar{q} = \frac{k_x}{k_v \Omega_0} q & \bar{r} = \frac{k_x}{k_v \Omega_0} r \\ \bar{\phi} = \phi & \bar{\theta} = \theta & \bar{\psi} = \psi \\ \bar{x} = \frac{1}{k_x R} x & \bar{y} = \frac{1}{k_x R} y & \bar{h} = \frac{1}{k_x R} h \\ \bar{\Omega} = \frac{\Omega}{\Omega_0} & \bar{C}_T = \frac{C_T}{k_T} & \bar{C}_i = \frac{C_i}{k_T} \\ \bar{\theta}_0 = \theta_0 & \bar{\theta}_c = \theta_c & \bar{\theta}_s = \theta_s \\ \bar{\theta}_t = \theta_t & \tau = \frac{k_v \Omega_0}{k_x} t & \bar{u}_0 = \frac{k_x}{k_v \Omega_0} u_0 \\ \bar{u}_c = \frac{k_x}{k_v \Omega_0} u_c & \bar{u}_s = \frac{k_x}{k_v \Omega_0} u_s & \bar{u}_t = \frac{k_x}{k_v \Omega_0} u_t \\ \bar{P}_S = \frac{k_x}{k_v \Omega_0^3 (I_R + k^2 I_{IR})} P_S \end{cases}$$

where k_x , k_v , and k_T are constant scaling factors. Thus the normalized and scaled optimal control problem is readily derived.

This optimal control problem has the prospect of singular arcs, which is often difficult to solve. To overcome this shortcoming and reflect a pilot's realistic control activity,^{12,13} the cost function is augmented with a modifying term to yield

$$\min J = J^p + \frac{1}{\tau_f - \tau_0} \int_{\tau_0}^{\tau_f} (w_1 \cdot \bar{u}_0^2 + w_2 \cdot \bar{u}_c^2 + w_3 \cdot \bar{u}_s^2 + w_4 \cdot \bar{u}_t^2) d\tau$$

where w_1 , w_2 , w_3 , and w_4 are the weighting functions associated with control rates, usually taken as constants.

4. Numerical solution techniques

The normalized and scaled optimal control problem is expressed as the following general continuous Bolza problem for convenience in this section. Determine the control vector, $\mathbf{u}(t)$, that minimizes the Bolza cost function

$$\min J = \Phi(\mathbf{x}(t_0), t_0, \mathbf{x}(t_f), t_f) + \frac{1}{t_f - t_0} \cdot \int_{t_0}^{t_f} g(\mathbf{x}(t), \mathbf{u}(t), t) dt$$

involving the state vector, $\mathbf{x}(t)$, the fixed initial time, t_0 , the free final time, t_f , subject to the state equations,

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t), \quad t \in [t_0, t_f]$$

the boundary conditions,

$$\boldsymbol{\varphi}(\mathbf{x}(t_0), t_0, \mathbf{x}(t_f), t_f) \begin{cases} = \mathbf{0} \\ \leq \mathbf{0} \end{cases}$$

and the path constraints,

$$\mathbf{C}(\mathbf{x}(t), \mathbf{u}(t), t) \begin{cases} = \mathbf{0} \\ \leq \mathbf{0} \end{cases}, t \in [t_0, t_f]$$

In this paper, the direct transcription approach is employed to solve the nonlinear optimal control problem.^{19,20} This technique allows the optimal control problem to be transcribed into a nonlinear programming problem (NLP). The basic idea behind direct transcription involves discretizing the state and control representation of a continuous trajectory.

The solution time interval $[t_0, t_f]$ of the optimal control problem is first divided into $N - 1$ segments so that

$$t_0 = t_1 < t_2 < \dots < t_k < \dots < t_N = t_f$$

Then for $k = 1, 2, \dots, N - 1$, the node points can be defined as

$$t_{k+1} = t_k + \Delta t_k$$

In case of equally spaced division points,

$$\Delta t_k = (t_f - t_0)/(N - 1)$$

In the direct transcription approach, the Hermite–Simpson collocation method is used to discretize the state equations. The corresponding NLP variables are

$$\mathbf{X} = [(\mathbf{x}, \mathbf{u}, \mathbf{u}_m)_1 \quad (\mathbf{x}, \mathbf{u}, \mathbf{u}_m)_2 \quad \dots \quad (\mathbf{x}, \mathbf{u}, \mathbf{u}_m)_{N-1} \quad (\mathbf{x}, \mathbf{u})_N \quad t_f]$$

with

$$\mathbf{x}_k = \mathbf{x}(t_k), \quad \mathbf{u}_k = \mathbf{u}(t_k), \quad \mathbf{u}_{m_k} = \mathbf{u}(t_{m_k}), \quad t_{m_k} = (t_k + t_{k+1})/2$$

The state equations of the optimal control problem are represented by a set of defect equality constraints as follows

$$\begin{aligned} \mathbf{x}_{k+1} - \mathbf{x}_k - \frac{\Delta t_k}{6} [\mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, t_k) + 4\mathbf{f}(\mathbf{x}_{m_k}, \mathbf{u}_{m_k}, t_{m_k}) \\ + \mathbf{f}(\mathbf{x}_{k+1}, \mathbf{u}_{k+1}, t_{k+1})] = \mathbf{0} \end{aligned} \quad (4)$$

where the state vector at the midpoint of a segment $[t_k, t_{k+1}]$ is given by

$$\mathbf{x}_{m_k} = \frac{1}{2} (\mathbf{x}_k + \mathbf{x}_{k+1}) + \frac{\Delta t_k}{8} [\mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, t_k) - \mathbf{f}(\mathbf{x}_{k+1}, \mathbf{u}_{k+1}, t_{k+1})]$$

The boundary conditions are imposed at the initial and terminal node points as follows

$$\boldsymbol{\varphi}(\mathbf{x}_1, \mathbf{x}_N, t_0, t_f) \begin{cases} = \mathbf{0} \\ \leq \mathbf{0} \end{cases} \quad (5)$$

The path constraints are enforced at all the node points and midpoints of each trajectory segment as follows:

$$\begin{cases} C(\mathbf{x}_k, \mathbf{u}_k, t_k) \Big\{ \begin{matrix} = 0 \\ \leq 0 \end{matrix} \\ C(\mathbf{x}_{m_k}, \mathbf{u}_{m_k}, t_{m_k}) \Big\{ \begin{matrix} = 0 \\ \leq 0 \end{matrix} \end{cases} \quad (6)$$

Eqs. (4)–(6) comprise the algebraic equality and inequality constraints of the NLP. As a result of the transcription, the cost function of the optimal control problem is replaced by the performance index of the NLP

$$\min I = \Phi(\mathbf{x}_1, \mathbf{x}_N, t_0, t_f) + \frac{1}{6} \sum_{k=1}^{N-1} \frac{1}{N-1} [g(\mathbf{x}_k, \mathbf{u}_k, t_k) + 4g(\mathbf{x}_{m_k}, \mathbf{u}_{m_k}, t_{m_k}) + g(\mathbf{x}_{k+1}, \mathbf{u}_{k+1}, t_{k+1})]$$

Now the original optimal control problem is approximated by the resulting NLP. The optimal solution $\mathbf{X}^*$ of the NLP can be obtained by solving the NLP numerically using a sparse sequential quadratic programming algorithm. The optimal solution of the optimal control problem can then be approximated from the information available at the solution $\mathbf{X}^*$ of the NLP. Specifically, for the state variables, piecewise cubic Hermite interpolation of the discrete solution is used to construct an approximation to the continuous solution by enforcing the approximation to match the state at the node points and the derivative of the approximation to match the derivative of the state at the node points. Besides, for the control variables, piecewise parabolic interpolation of the discrete solution is used to construct an approximation to the continuous solution from the values at the node points as well as the values of the control at the midpoints of each segment.

5. Results and discussion

A UH-60 helicopter is taken as the sample helicopter in this work. To be compared with the optimal results computed using a two-dimensional point-mass model in Ref. ⁵, the optimal control inputs and autorotation landing trajectory are first calculated with the cost function J_1 instead of J in the optimal control problem. Here J_1 is given by

$$\min J_1 = w_x \cdot \bar{x}_f + \frac{1}{\tau_f - \tau_0} \times \int_{\tau_0}^{\tau_f} (w_1 \cdot \bar{u}_0^2 + w_2 \cdot \bar{u}_c^2 + w_3 \cdot \bar{u}_s^2 + w_4 \cdot \bar{u}_t^2) d\tau$$

where w_x is a weighting factor corresponding to the horizontal landing distance x_f .

Fig. 1 compares the time histories of the optimal solutions calculated using the six-degree-of-freedom rigid-body model and a two-dimensional point-mass model⁵ for UH-60 helicopters with a gross weight of 8618 kg. Fig. 2 shows the additional time histories of the optimal solutions obtained using the six-degree-of-freedom rigid-body model. The OEI initial conditions are $h_0 = 38.1$ m, $\gamma_0 = -3^\circ$, and $V_0 = 16.76$ m/s, and the safe touchdown vertical and horizontal speed limits are 1.524 m/s and 12.192 m/s, respectively. The engine failure is assumed to happen at $t = 0$ s, and no time delay is applied, i.e., $t_0 = 0$ s. The weighting factors are chosen as follows: $w_1 = 2.0$, $w_2 = 2.0$, $w_3 = 1.0$, $w_4 = 1.0$, and $w_x = 0.2$.

The comparison shown in Fig. 1 is reasonably good. There is obvious difference between the time histories of the thrust

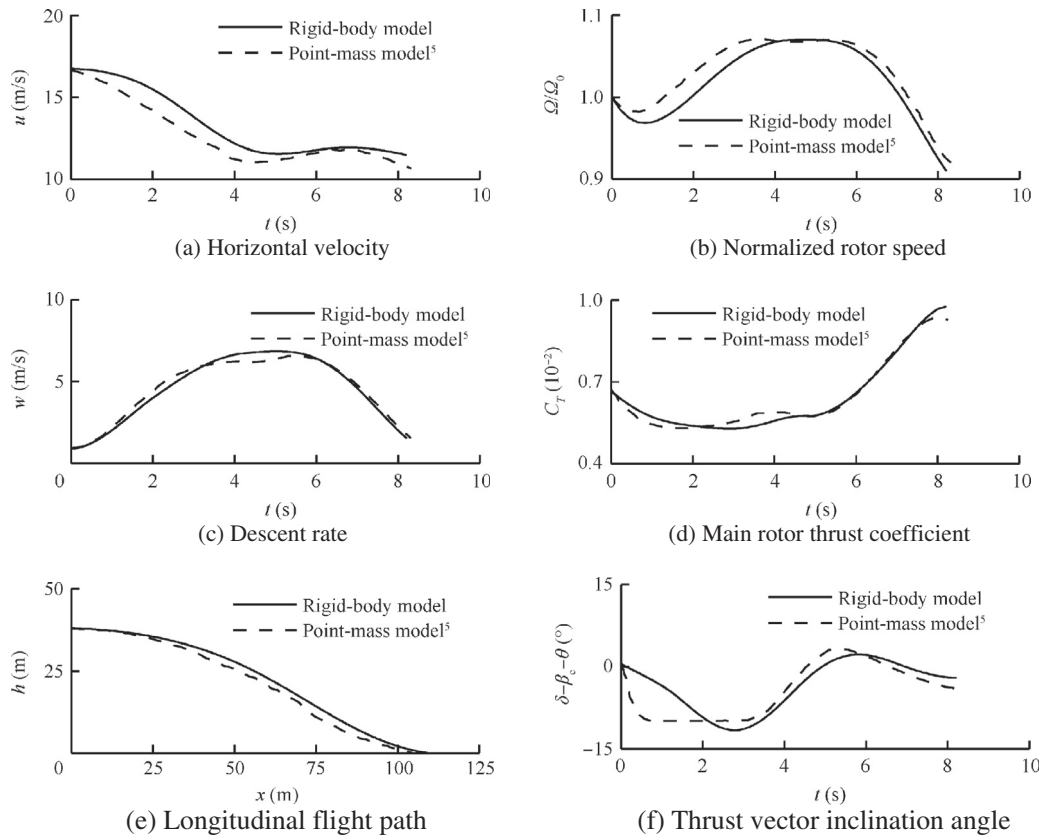


Fig. 1 Comparison of the optimal solutions using the rigid-body model and the point-mass model.⁵

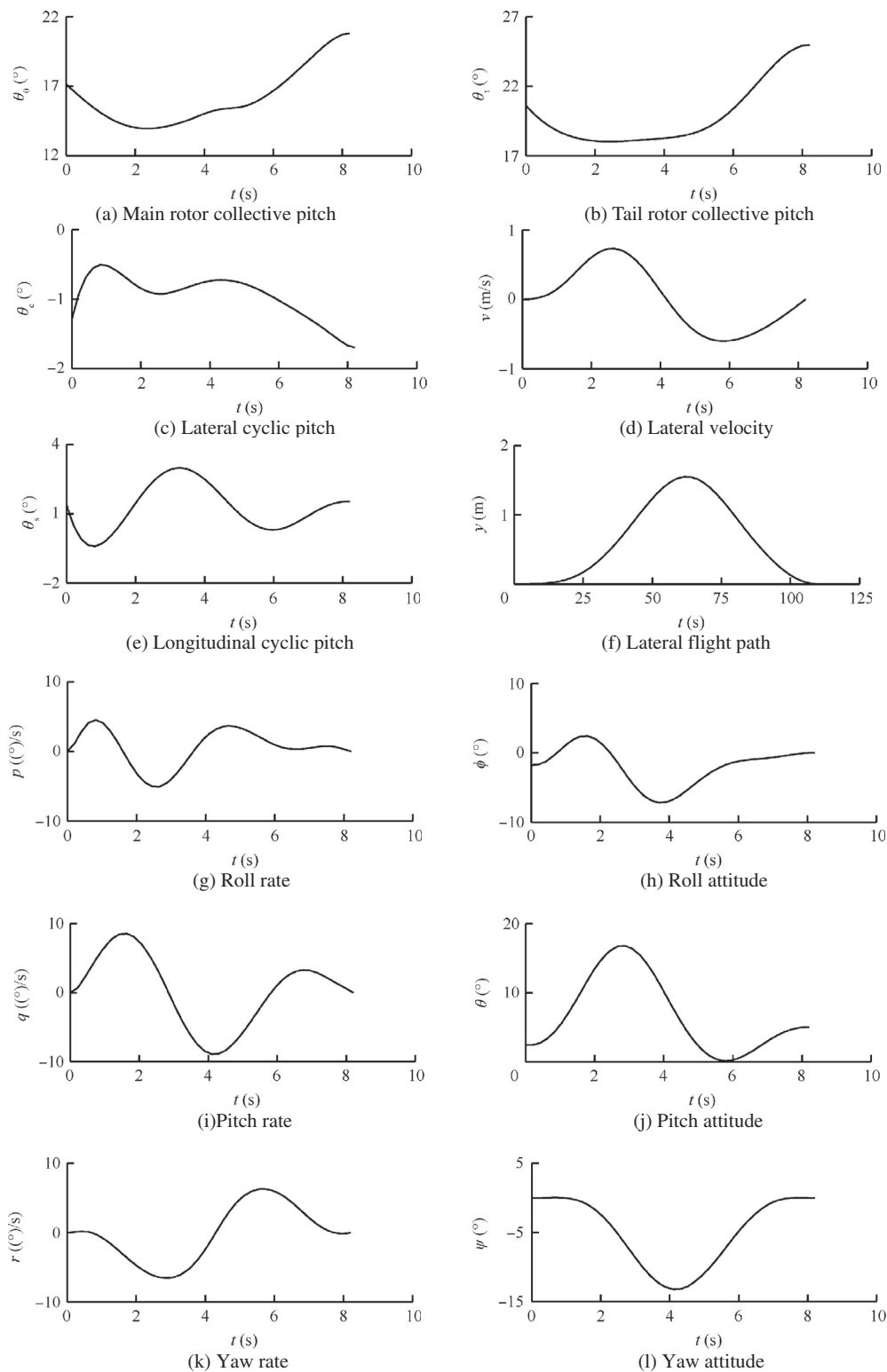


Fig. 2 Additional time histories of the optimal solutions using the rigid-body model.

vector inclination angle immediately following engine failure. This difference is largely due to the absence of equations and constraints on the pitch angular motion in the point-mass model, and the fact that the thrust vector inclination angle is primarily produced by the pitch attitude. As a result, for the point-mass model, the thrust vector is tilted backward to its

limit more rapidly to decelerate the aircraft, and therefore more kinetic energy is utilized to replenish the rotational energy in the rotor. In addition to those time histories used to compare with the optimal solutions using the point-mass model, the optimal solutions using the rigid-body model furthermore involve the time histories of angular rates and

attitudes, lateral velocity and position, as well as pitch controls, as depicted in Fig. 2. In summary, the rigid-body model can produce more satisfactory results.

The optimal control strategy shown in Fig. 2, which leads to hard touchdown with the maximum safe touchdown descent rate, differs from the autorotation technique employed in flight tests. In most of the flight tests, a minimum time delay of 1 s between engine failure and control actuation is required, and pilots are normally instructed to obtain a minimum touchdown speed. To reflect this piloting technique, the optimal con-

trol problem with the cost function J is solved, in which 1-s time delay is taken into account. However, it is observed that, for some cases of problems, the optimal control inputs are hardly realistic, the optimal computed values of t_f and x_f are often large, and the touchdown site in the optimal trajectory usually does not correspond to the final time instant t_f .²¹ To address this issue, the optimal control problem can be reformulated by modifying the cost function in two approaches. One approach is to introduce the elapsed time ($t_f - t_0$) during the optimal control flight in the cost function as follows:

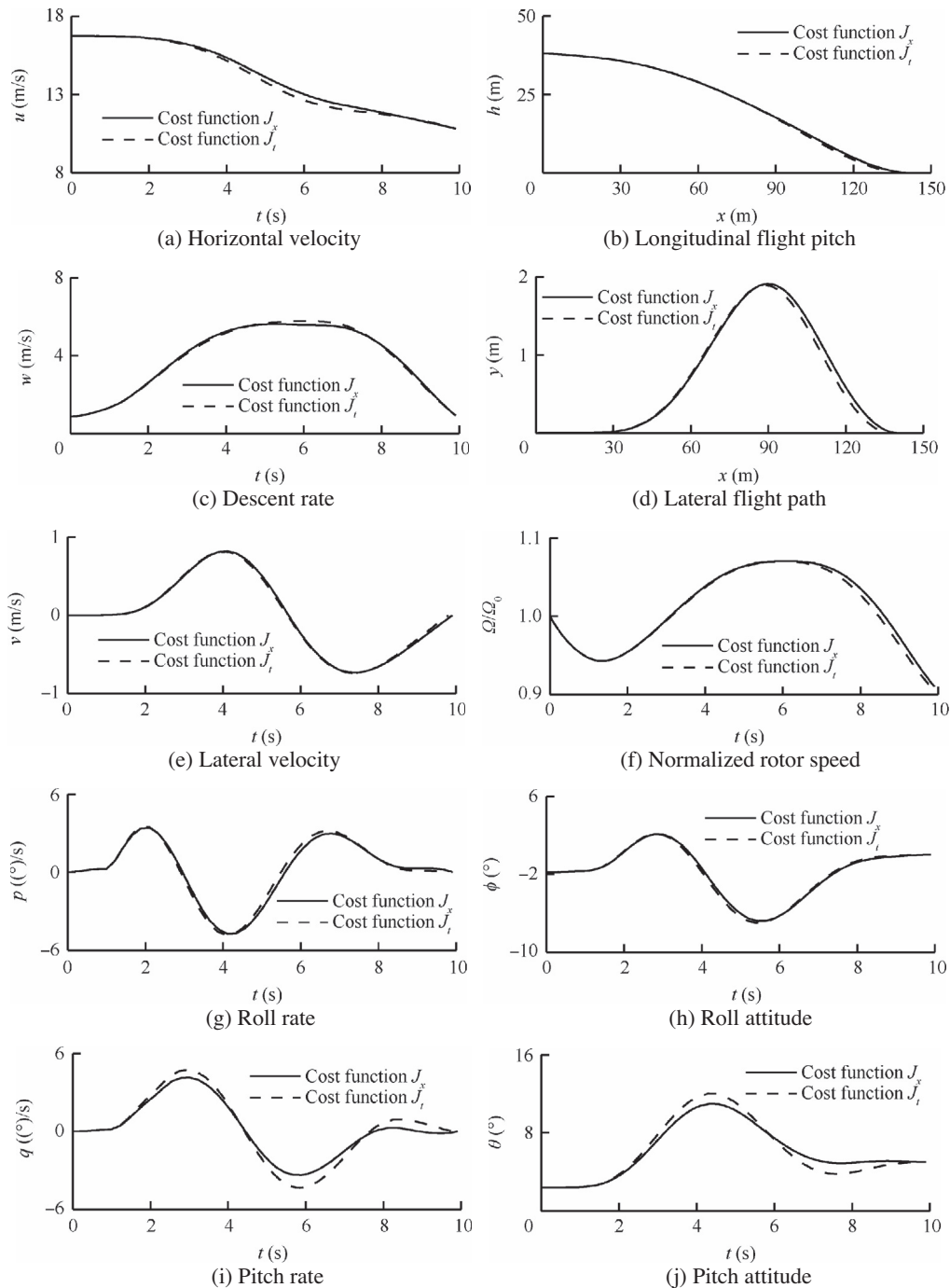


Fig. 3 Comparison of the optimal solutions with the cost functions J_t and J_x .

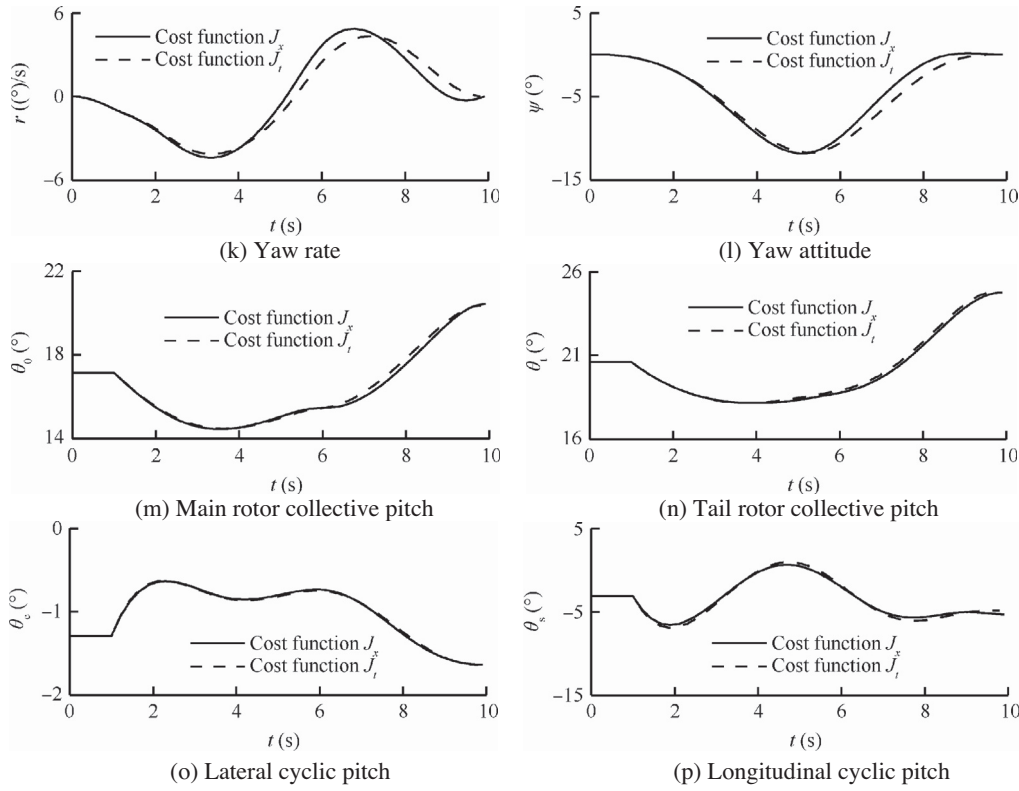


Fig. 3 (continued)

$$\min J_t = w_v \cdot \left[\left(\frac{u(t_f)}{u_{fmax}} \right)^2 + \left(\frac{w(t_f)}{w_{fmax}} \right)^2 \right] + w_t \cdot (\tau_f - \tau_0) + \frac{1}{\tau_f - \tau_0} \times \int_{\tau_0}^{\tau_f} (w_1 \cdot \bar{u}_0^2 + w_2 \cdot \bar{u}_c^2 + w_3 \cdot \bar{u}_s^2 + w_4 \cdot \bar{u}_t^2) d\tau$$

where w_v is a weighting factor associated with the touchdown speed and w_t is a weighting factor corresponding to the optimal control flight time. Alternatively, one could introduce the horizontal landing distance x_f in the cost function as follows

$$\min J_x = w_v \cdot \left[\left(\frac{u(t_f)}{u_{fmax}} \right)^2 + \left(\frac{w(t_f)}{w_{fmax}} \right)^2 \right] + w_x \cdot \bar{x}_f + \frac{1}{\tau_f - \tau_0} \times \int_{\tau_0}^{\tau_f} (w_1 \cdot \bar{u}_0^2 + w_2 \cdot \bar{u}_c^2 + w_3 \cdot \bar{u}_s^2 + w_4 \cdot \bar{u}_t^2) d\tau$$

where w_x is a weighting factor corresponding to the horizontal landing distance x_f . The weighting factors in the cost function are used to adjust the relative importance of the corresponding terms with respect to the others.

Fig. 3 compares the time histories of the optimal solutions calculated with the cost function J_t and J_x for a UH-60 helicopter with a gross weight of 8618 kg. The OEI initial conditions are $h_0 = 38.1$ m, $\gamma_0 = -3^\circ$, and $V_0 = 16.76$ m/s, and the safe touchdown vertical and horizontal speed limits are 1.524 m/s and 12.192 m/s, respectively. The engine failure is assumed to happen at $t = 0$ s, and 1-s time delay is applied, i.e., $t_0 = 1$ s. The weighting factors are chosen as follows: $w_1 = 4.0$, $w_2 = 4.0$, $w_3 = 2.0$, $w_4 = 1.0$, $w_v = 0.015$,

$w_t = 0.1$ in the cost function J_t , and $w_1 = 2.5$, $w_2 = 2.5$, $w_3 = 2.0$, $w_4 = 1.0$, $w_v = 0.015$, $w_x = 0.1$ in the cost function J_x .

The optimal solutions calculated with the cost functions J_t and J_x are in fairly good agreement. For the optimal control strategies obtained with the cost functions J_t and J_x , the rates of descent are reduced to 1.106 m/s and 1.298 m/s at touchdown, while at the same time the horizontal speeds remain at 10.924 m/s and 9.851 m/s, respectively. This work did not try to reduce the landing speed further, considering the feasibility of the control strategies. The optimal control formulation with the cost function either J_t or J_x is able to produce reasonable control strategies and trajectories.

6. Conclusions

- (1) The optimal control method has been applied successfully to an augmented six-degree-of-freedom rigid-body flight dynamic model to study the safe control strategy and autorotation landing trajectory for a twin-engine helicopter after one-engine failure.
- (2) The optimal solutions obtained using the six-degree-of-freedom rigid-body model and a two-dimensional point-mass model show reasonably good agreement. In addition, the optimal solutions using the six-degree-of-freedom rigid-body model furthermore involve the time histories of angular rates and attitudes, lateral velocity and position, as well as pitch controls.

- (3) Taking account of 1-s time delay and minimum touch-down speed, the optimal control formulations proposed in this paper are able to produce reasonable control strategies and trajectories.

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